

1. Prove that,

(i) $P_n(1) = 1$

(ii) $P_n(-1) = (-1)^n$

(iii) $P_n(-x) = (-1)^n P_n(x)$

Using the generating function for Legendre polynomials,

$$(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x) z^n \quad |x| \leq 1, |z| < 1$$

(i) $P_n(1) = 1$

Setting $x = 1$

$$(1-2z+z^2)^{-\frac{1}{2}} = (1-z)^{-1} = \sum_{n=0}^{\infty} P_n(1) z^n$$

$$\Rightarrow \frac{1}{1-z} = \sum_{n=0}^{\infty} P_n(1) z^n$$

Since,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \text{ we can equate the coefficients of } z^n, \\ \text{we get, } P_n(1) = 1 \quad (\text{Proved})$$

(ii) $P_n(-1) = (-1)^n$

Let, $x = -1$ in generating function;

$$\frac{1}{\sqrt{1+2z+z^2}} = \sum_{n=0}^{\infty} P_n(-1) z^n$$

$$\Rightarrow \frac{1}{1+z} = \sum_{n=0}^{\infty} P_n(-1) z^n$$

Since,

$$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n, \text{ we can equate the coefficients of } z^n, \\ \text{to get, } P_n(-1) = (-1)^n \quad (\text{Proved})$$

(iii) $P_n(-x) = (-1)^n P_n(x)$

Replacing x with $-x$ in generating function:

$$\frac{1}{\sqrt{1+2xz+z^2}} = \sum_{n=0}^{\infty} P_n(-x) z^n$$

Now,

Replacing z with $-z$:

$$\frac{1}{\sqrt{1+2xz+z^2}} = \sum_{n=0}^{\infty} P_n(-x) (-1)^n z^n$$

We also have,

$$\frac{1}{\sqrt{1+2xz+z^2}} = \sum_{n=0}^{\infty} P_n(x) (-z)^n = \sum_{n=0}^{\infty} (-1)^n z^n P_n(x)$$

Equating the two expressions:

$$\sum_{n=0}^{\infty} z^n P_n(-x) = \sum_{n=0}^{\infty} (-1)^n z^n P_n(x)$$

Equating coefficients of z^n :

$$P_n(-x) = (-1)^n P_n(x)$$

Hence,

$$P_n(-x) = (-1)^n P_n(x) \quad (\text{Proved})$$

2 □ Prove that $(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} z^n P_n(x)$, $|x| \leq 1$, $|z| < 1$

Let, $u = 2xz - z^2$

The expression becomes: $(1-u)^{-\frac{1}{2}}$

Using binomial expansion:

$$(1-u)^{-\frac{1}{2}} = 1 + \frac{u}{2} + \frac{1 \times 3}{2 \times 4} u^2 + \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}$$

We can write,

$$(1-(2xz-z^2))^{-\frac{1}{2}} = \sum_{k=0}^{\infty} \frac{(2k)!}{2^{2k} (k!)^2} (2xz-z^2)^k$$

Expanding,

$(2xz-z^2)^k$ using binomial expansion:

$$\begin{aligned} (2xz-z^2)^k &= \sum_{m=0}^k \binom{k}{m} (2xz)^{k-m} (-z^2)^m \\ &= \sum_{m=0}^k \frac{k!}{m!(k-m)!} (2x)^{k-m} z^{k-m} (-1)^m z^{2m} \\ &= \sum_{m=0}^k \frac{k! (-1)^m (2x)^{k-m}}{m!(k-m)!} z^{k+m} \end{aligned}$$

Combining the two summations:

$$(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{k=0}^{\infty} \sum_{m=0}^k \frac{(2k)!}{2^{2k} (k!)^2} \frac{k! (-1)^m (2x)^{k-m}}{m!(k-m)!} z^{k+m}$$

To find coefficients of z^n , let $n = k+m$. This implies $k = n-m$, as m must be $\leq k$, we have $m \leq n-m$ or $2m \leq n$; Thus m goes from 0 to $\frac{n}{2}$

The coefficient of z^n in the double sum is:

$$\sum_{m=0}^{n/2} \frac{(2n-2m)!}{2^{2n-2m} (n-m)! (n-2m)! m!} (-1)^m (2x)^{n-2m}$$

Simplifying constants:

$$\sum_{m=0}^{\lfloor \frac{n}{2} \rfloor} \frac{(-1)^m (2n-2m)!}{2^n m! (n-m)! (n-2m)!} x^{n-2m}$$

The expression above is the exact series definition of the Legendre Polynomial $P_n(x)$. Therefore:

$$(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} z^n P_n(x) \quad (\text{Proved})$$

3 $\square$ Show that $1 + \frac{1}{2} P_1(\cos \theta) + \frac{1}{3} P_2(\cos \theta) + \dots = \log \left(\frac{1 + \sin \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right)$

The generating function for Legendre polynomial,

$$\frac{1}{\sqrt{1-2xz+z^2}} = \sum_{n=0}^{\infty} P_n(x) z^n$$

Substituting $x = \cos \theta$ and integrating with respect to z ,

$$\frac{1}{\sqrt{1-2\cos \theta z+z^2}} = \sum_{n=0}^{\infty} P_n(\cos \theta) z^n$$

Integrating both sides with respect to z from 0 to z ,

$$\int_0^z \frac{dz}{\sqrt{1-2z\cos \theta+z^2}} = \int_0^z \sum_{n=0}^{\infty} P_n(\cos \theta) z^n dz$$

$$\Rightarrow \int_0^z \frac{dz}{\sqrt{1-2z\cos\theta+z^2}} = \sum_{n=0}^{\infty} P_n(\cos\theta) \int_0^z z^n dz$$

$$\Rightarrow \int_0^z \frac{dz}{\sqrt{1-2z\cos\theta+z^2}} = \sum_{n=0}^{\infty} \frac{P_n(\cos\theta) z^{n+1}}{n+1}$$

Here, ($z=1$)

$$\begin{aligned} \int_0^z \frac{dz}{\sqrt{1-2z\cos\theta+z^2}} &= \log(z - \cos\theta + \sqrt{1-2z\cos\theta+z^2}) \Big|_0^z \\ &= \log(1 - \cos\theta + \sqrt{1-2\cos\theta+2^2}) - (-\log(1)) \\ &= \log(1 - \cos\theta + \sqrt{1-2\cos\theta+2^2}) \end{aligned}$$

Let, $z=1$

$$\begin{aligned} &= \log(1 - \cos\theta + \sqrt{2-2\cos\theta}) - \log(2\sin^2\frac{\theta}{2}) \\ &= \log(1 - \cos\theta + \sqrt{2(1-\cos\theta)}) - \log(2\sin^2\frac{\theta}{2}) \\ &= \log(1 - \cos\theta + \sqrt{2 \cdot 2\sin^2\frac{\theta}{2}}) \\ &= \log(2\sin^2\frac{\theta}{2} + 2|\sin\frac{\theta}{2}|) - \log(2\sin^2\frac{\theta}{2}) \\ &= \log[2\sin\frac{\theta}{2}(\sin\frac{\theta}{2}+1)] - \log(2\sin^2\frac{\theta}{2}) \\ &= \log\left(\frac{2\sin\frac{\theta}{2}(\sin\frac{\theta}{2}+1)}{2\sin^2\frac{\theta}{2}}\right) \\ &= \log\left(\frac{1+\sin\frac{\theta}{2}}{\sin\frac{\theta}{2}}\right) \end{aligned}$$

Substituting z in the series:

$$\sum_{n=0}^{\infty} \frac{P_n(\cos\theta)}{n+1} = 1 + \frac{1}{2} P_2(\cos\theta) + \frac{1}{3} P_3(\cos\theta) + \dots$$

(Proved)

4 □ From orthogonal properties of Legendre's polynomials, prove that, $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$, where

$$\delta_{mn} = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases}$$

Legendre polynomials $P_n(x)$ satisfy the Legendre differential equation:

$$\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} P_n(x) \right] + n(n+1) P_n(x) = 0$$

Similarly for $P_m(x)$

$$\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} P_m(x) \right] + m(m+1) P_m(x) = 0$$

Case: 1 $m \neq n$ (Orthogonality)

Multiply the first equation by P_m and second by P_n , subtract and integrate over $[-1, 1]$

$$[n(n+1) - m(m+1)] \int_{-1}^1 P_m P_n dx = \left[(1-x^2) (P_m P_n' - P_n P_m') \right]_{-1}^1 = 0$$

Since, $m \neq n$ implies $n(n+1) \neq m(m+1)$ we conclude:

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0 \quad (m \neq n)$$

Case: 2 $m = n$

We use the generating function $(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x) z^n$ and square both sides, then

$$\int_{-1}^1 (1-2xz+z^2)^{-1} dx = \sum \left[\int_{-1}^1 P_m P_n dx \right] z^{m+n}$$

The left side corresponds to

$$\begin{aligned} & \left[-\frac{1}{2z} \log(1-2xz+z^2) \right]_{-1}^1 \\ &= \frac{1}{2} \log \left| \frac{1+z}{1-z} \right| \\ &= \frac{1}{2} 2 \sum_{n=0}^{\infty} \frac{z^{2n+1}}{2n+1} \\ &= \sum_{n=0}^{\infty} \frac{2}{2n+1} z^{2n} \end{aligned}$$

Comparing coefficients of z^n ,

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

Combining both cases;

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn} \quad (\text{Proved})$$

□ Prove that $\frac{1-z^2}{(1-2xz+z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) z^n P_n(x)$

$$\text{Let, } G(x, z) = (1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x) z^n$$

Differentiating G with respect to z :

$$\begin{aligned} \frac{\delta G}{\delta z} &= -\frac{1}{2} (1-2xz+z^2)^{-\frac{3}{2}} (-2x+2z) \\ &= \frac{x-z}{(1-2xz+z^2)^{3/2}} \end{aligned}$$

Also, from the series,

$$\frac{\delta G}{\delta z} = \sum_{n=1}^{\infty} n P_n(x) z^{n-1}$$

Now,

$$(1-2xz+z^2) \frac{\delta G}{\delta z} = (x-z) \cdot G(x, z)$$

$$\sum_{n=0}^{\infty} n P_n x^{n-1} - 2x \sum n P_n z^n + \sum n P_n z^{n+1}$$

$$= x \sum P_n z^n - \sum P_n z^{n+1}$$

Multiplying G by z , and differentiating,

$$\frac{d}{dz} (z \cdot G) = G + z \cdot \frac{\delta G}{\delta z} \text{ leading to ...}$$

$$\frac{d}{dz} (z \cdot G) = \sum_{n=0}^{\infty} (n+1) P_n z^n$$

Differentiating the identity:

$$G^2 = \frac{1}{(1-2xz+z^2)}$$

$$2G \frac{\delta G}{\delta z} = \frac{2(x-z)}{(1-2xz+z^2)} \dots$$

We know, the recurrence relation:

$$(n+1) P_{n+1}(x) - (2n+1)x P_n(x) + n P_{n-1}(x) = 0$$

Multiplying $G(x, z)$ by $(1-z^2)$

$$\begin{aligned} (1-z^2) G &= (1-z^2) \sum P_n z^n \\ &= \sum P_n z^n - \sum P_n z^{n+2} \end{aligned}$$

From the generating function,

$$(1-2xz+z^2)^{\frac{3}{2}} = \sum (2n+1) P_n z^n$$

We differentiate G with respect to z and use,

$$(1-2xz+z^2) \frac{\partial G}{\partial z} = (x-z)G$$

Let $x \cdot \frac{\partial G}{\partial z} = \sum n P_n z^n$, $\frac{\partial(zG)}{\partial z} = \sum (n+1) P_n z^n$

Thus, $2z \cdot \frac{\partial G}{\partial z} + G = \sum (2n+1) P_n z^n$, multiplying both sides by $(1-2xz+z^2)^{\frac{1}{2}}$ using $(x-z)G = (1-2xz+z^2) \frac{\partial G}{\partial z}$ we can show,

$$(1-z)^2 (1-2xz+z^2)^{-\frac{3}{2}} = \sum_{n=0}^{\infty} (2n+1) P_n(x) z^n \quad (\text{Proved})$$

6 $\square$ Show that $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$, $n \geq 1$

The generating function:

$$G = (1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x) z^n$$

Differentiating both sides with respect to z :

$$\frac{\partial G}{\partial z} = (x-z)(1-2xz+z^2)^{-\frac{3}{2}} = \sum_{n=1}^{\infty} n P_n z^{n-1}$$

Multiplying by $(1-2xz+z^2)$

$$(x-z)G = (1-2xz+z^2) \sum n P_n z^{n-1}$$

Expanding the left side:

$$x \sum P_n z^n - \sum P_n z^{n+1}$$

Expanding the right side by distributing,

$$\sum n P_n z^{n-1} - 2x \sum n P_n z^n + \sum n P_n z^{n+1}$$

Let, $m=n-1$, in first sum; $m=n$ in second; $m=n+1$ in third,

$$\sum (m+1) P_{m+1} z^m - 2x \sum m P_m z^m + \sum (m-1) P_{m-1} z^m$$

Equating coefficients of x^n ,

$$(n+1) P_{n+1} - 2xn P_n + (n-1) P_{n-1} = x P_n - P_{n-1}$$

$$\therefore (n+1) P_{n+1} = (2n+1)x P_n - n P_{n-1}$$

(Proved)

7 $\square$ Prove that,

$$(i) J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

$$(ii) J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$(iii) (J_{1/2}(x))^2 + (J_{-1/2}(x))^2 = \frac{2}{\pi x}$$

(i) The Bessel function of first kind of order ν is defined by,

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\nu+1)} \left(\frac{x}{2}\right)^{2m+\nu}$$

for $\nu = -\frac{1}{2}$

$$J_{-\frac{1}{2}}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\frac{1}{2})} \left(\frac{x}{2}\right)^{2m-\frac{1}{2}}$$

$$J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\frac{1}{2})} \left(\frac{x}{2}\right)^{2m}$$

We know,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$\Gamma\left(m+\frac{1}{2}\right) = \frac{(2m)!}{4^m \cdot m!} \sqrt{\pi}$$

$$\begin{aligned} J_{-1/2}(x) &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m (2m)!}{4^m (m!)^2} x^{2m} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} x^{2m} \end{aligned}$$

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

(Proved)

(ii) $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$

$$v = \frac{1}{2}$$

$$J_{1/2}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma\left(m+\frac{3}{2}\right)} \left(\frac{x}{2}\right)^{2m+\frac{1}{2}}$$

$$J_{1/2}(x) = \sqrt{\frac{2}{x}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma\left(m+\frac{3}{2}\right)} \left(\frac{x}{2}\right)^{2m+1}$$

We know,

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

$$\Gamma\left(m+\frac{3}{2}\right) = \frac{(2m+1)!}{4^m m!} \frac{\sqrt{\pi}}{2}$$

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sum_{m=0}^{\infty} \frac{(-1)^m \cdot 2 \cdot 4^m \cdot m!}{(2m+1)!} x^{2m+1}$$

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

(Proved)

$$(iii) \left[J_{1/2}(x) \right]^2 + \left[J_{-1/2}(x) \right]^2 = \frac{2}{\pi x}$$

$$\left[J_{1/2}(x) \right] = \sqrt{\frac{2}{\pi x}} \sin x$$

$$\left[J_{-1/2}(x) \right] = \sqrt{\frac{2}{\pi x}} \cos x$$

$$\left[J_{1/2}(x) \right]^2 + \left[J_{-1/2}(x) \right]^2 = \frac{2}{\pi x} (\sin^2 x + \cos^2 x)$$

$$\therefore \left[J_{1/2}(x) \right]^2 + \left[J_{-1/2}(x) \right]^2 = \frac{2}{\pi x} \quad (\text{Proved})$$

8 □ For positive integer n , show that $J_{-n}(x) = (-1)^n J_n(x)$

The Bessel function of first kind;

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\nu+1)} \left(\frac{x}{2}\right)^{2m+\nu}$$

replacing ν with $-n$ where n is a positive integer,

$$J_{-n}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m-n+1)} \left(\frac{x}{2}\right)^{2m-n}$$

Let, $m = k+n$

$$J_{-n}(x) = \sum_{k=-n}^{\infty} \frac{(-1)^{k+n}}{(k+n)! \Gamma(k+1)} \left(\frac{x}{2}\right)^{2(k+n)-n}$$

Since, $\Gamma(k+1)$ is infinite for $k < 0$, the summation starts from $k=0$

$$J_{-n}(x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+n}}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k+n}$$

$$J_{-n}(x) = (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+n+1)} \left(\frac{x}{2}\right)^{2k+n}$$

$$\text{So, } J_{-n}(x) = (-1)^n J_n(x) \quad (\text{Proved})$$